EXERCISE 6

Advanced SQL: DATE Functions

Question 1. Add 6 Months to each employee's hire date using DATEADD().

```
SELECT emp_id,
       name,
       hire_date
       DATEADD(month, 6, hire_date) As hire_plus_6_Months
FROM Employees;
```

Table 1: Employees.

emp_id	name	hire_date	hire_plus_6_Months
1	Alice	2020-01-15	2020-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

Question 2. Use DATEDIFF() to find age in days from dob to today.

```
SELECT student_id,
       name,
       DATEDIFF(day, dob, GETDATE()) As age_in_days
FROM students;
```

Table 2: Students

student_id	name	age_in_days
101	Maya	6858
102	Ethan	7110
103	Sienna	6641

Question 3: Find how many days are left until each event using DATEDIFF

```
SELECT event_id,
       event_name
       DATEDIFF (day, GETDATE( ), event_date) AS days_remaining
       FROM Events;
```

Table 3: Events.

event_id	event_name	day-remaining
1	Seminar	26
2	Workshop	469
3	Hackathon	245

Question 4. Calculate the number of days between issue_date and due_date

```
SELECT invoice_id,
       issue_date,
       due_date,
DATEDIFF (due_date, issue_date) AS days_between
FROM Invoices;
```

Table 4: Invoices

Invoice_id	issue_date	due_date	days-between
501	2025-03-10	2025-03-25	15
502	2025-04-01	2025-04-15	14
503	2025-04-10	2025-04-20	10

Question 5: Format start_date as 'Month YYYY' using TO_CHAR().

~~SELECT~~

```
SELECT Course_id,
       name,
       TO_CHAR(start_date, 'Month YYYY') As Formated_date
       FROM Course;
```

Table 5: Courses

Course_id	name	Formatted_date
201	SQL basics	May-2025
202	Python	June 2025

Question 6 Create full date from parts using DATE_FROM_PARTS()

```
SELECT Member_id,
 DATE_FROM_PARTS(start_year, start_month, start_day)
 AS Full_start_date
FROM Memberships;
```

Table 6: Membership

Member_id	full_start_date
1	2023-05-10
2	2022-11-25

Question 7: Extend each renewal_date by 1 using DATEADD()

```
SELECT sub_id,
       plan,
       DATEADD(year, 1, renewal_date) AS extended_renewal
        date
       FROM Subsuphón;
```

Table 7: Subsuphons

Sub_id	plan	extended-renewal_date
11	Basic	2026-01-01
12	Premium	2026-03-15

Question 8: Show current date and difference from order_date.
Use CURRENT_DATE and DATEDIFF().

```
SELECT Order_id,
       Order_date,
       ~~Current_Da~~
       CURRENT_DATE AS TODAY_date,
       DATEDIFF(CURRENT_DATE, order_date) AS days-since-
        order
       From orders;
```

Table 8: Orders.

Order_id	order_date	today_date	days-since-order
1001	2025-04-15	2026-04-21	371
1002	2025-04-10	2026-04-21	376

Question 9: Extract the year from training_date using DATE_PART() or EXTRACT()

```
SELECT training_id,
       topic,
       EXTRACT(YEAR FROM training_date) As training_year
FROM Trainings;
```

Table 9: Trainings

training_id	topic	training_year
1	Safety	2025
2	Compliance	2025

Question 10. Extract hour and minute from published_on

```
SELECT post_id,
       title
Extract (HOUR FROM published_on) As hour_published,
Extract (MINUTE FROM published_on) As minuted_published
FROM Blog_posts
```

Table 10: Blog_posts

post_id	title	hour_published	minute_published
1	SQL tips	10	15
2	Data cleaning	16	45

Question 11: Calculate days left until license expiry using DATEDIFF() and today's day

```
SELECT driver_id,
       license_expiry
DATEDIFF(day, GETDATE() license_expiry) As days_left FROM Drivers;
```

Table 11: Drivers

driver_id	lincense_expiry	days_left
301	2025-08-10	-255
302	2023-12-31	-843

Q12: Display the current timestamp and calculate seconds since the message was sent

SELECT Message_id,
Sent_timestamp,
~~Current~~
CURRENT_TIMESTAMP As current_timestamp,
DATEDIFF (SECONDS Sent_~~message~~ timestamp, CURRENT_TIMESTAMP) As Seconds_since_sent
FROM Messages;

Table 12: Messages

Message_id	Sent_timestamp	current_timestamp	seconds_since_sent
1	2025-04-19 09:32:45	2026-04-22 06:17:00	31,783,455
2	2025-04-18 23:59:59	2026-04-22 06:17:00	31,817,821

Q13. Add 15 days to return_date using DATEADD() to show restock_date

SELECT return_id,
return_date,
DATEADD(day,15,return_date) As restock_date
FROM Returns

Table 13: Returns

return_id	return_date	restock_date
901	2025-04-05	2025-04-20
902	2025-04-01	2025-04-16

Question 14: Convert assigned_on to date using TO_DATE() (If its stored as string).

```
SELECT  assigned_id,
        TO_DATE(assigned_on, 'YYYY-MM-DD') AS assigned_on_date
        FROM Assignments;
```

assign_id	assigned_on_date
1	2025-03-01
2	2025-03-05

Question 15: Convert scheduled_time to formatted string like 'April 19, 2025 at 2:00 pm' using TO_CHAR().

```
SELECT meeting_id,
   TO_CHAR(scheduled_time, 'month DD, YYYY "at" HH12:MI AM')
   AS formatted_meeting_time
FROM Meetings;
```

Meeting_id	Formatted_meeting_time
1	April 19, 2025 at 02:00 pm
2	April 19, 2025 at 09:30 AM